

Geodatabases and geographical information systems

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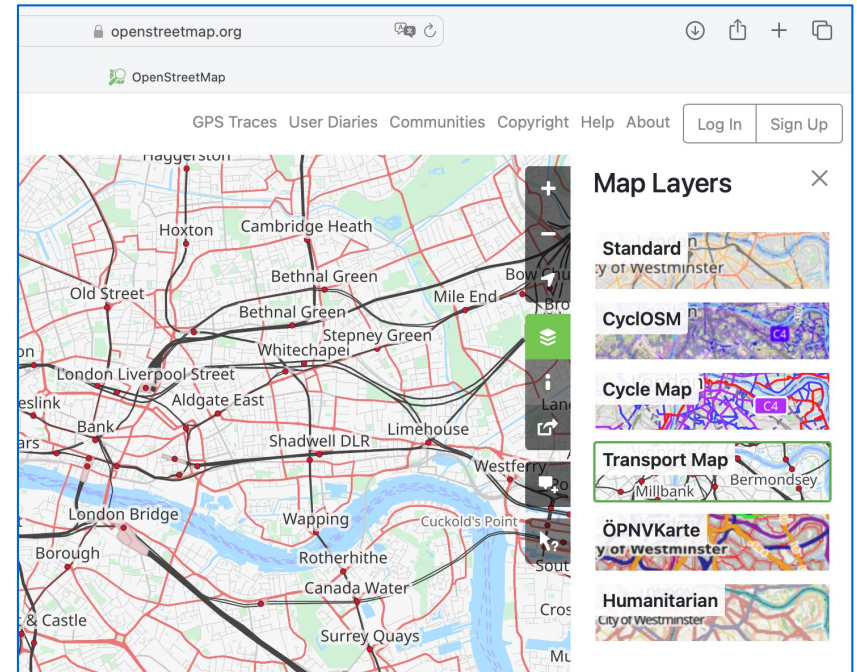
MSc program in Data Science has been developed
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Introduction

- Many real-life entities are described also with their location. This includes:
 - Employees working in the field
 - Transportation networks and vehicles
 - Buildings
 - Assets comprising on utility networks such as water supply networks
 - Police units
 - Public services such as schools, swimming pools, city hall offices,....
- In many cases, location of these entities should be described with geo-coordinates rather than addresses only. Examples include: fire hydrants located in parks, power grid lines located in forest areas,....
- In such cases, geodatabases a.k.a. spatial databases (*geobazy*, *przestrzenne bazy danych*) i.e. databases with extensive support for geographic data storage and processing are needed

Geodatabases

- Geodatabases are used to store records, while including their geographical coordinates and providing tools to process spatial data
- Typically, data are stored as numerical maps (*mapy numeryczne*), such as a map of the power grid network in the country or in one city only
- Maps are typically stored in the form of layers
- A layer (warstwa) is a collection of objects of the same category
- Examples include: a layer of rivers, a layer of buildings, a layer of address points, a layer of satellite images



Source: <https://www.openstreetmap.org/>

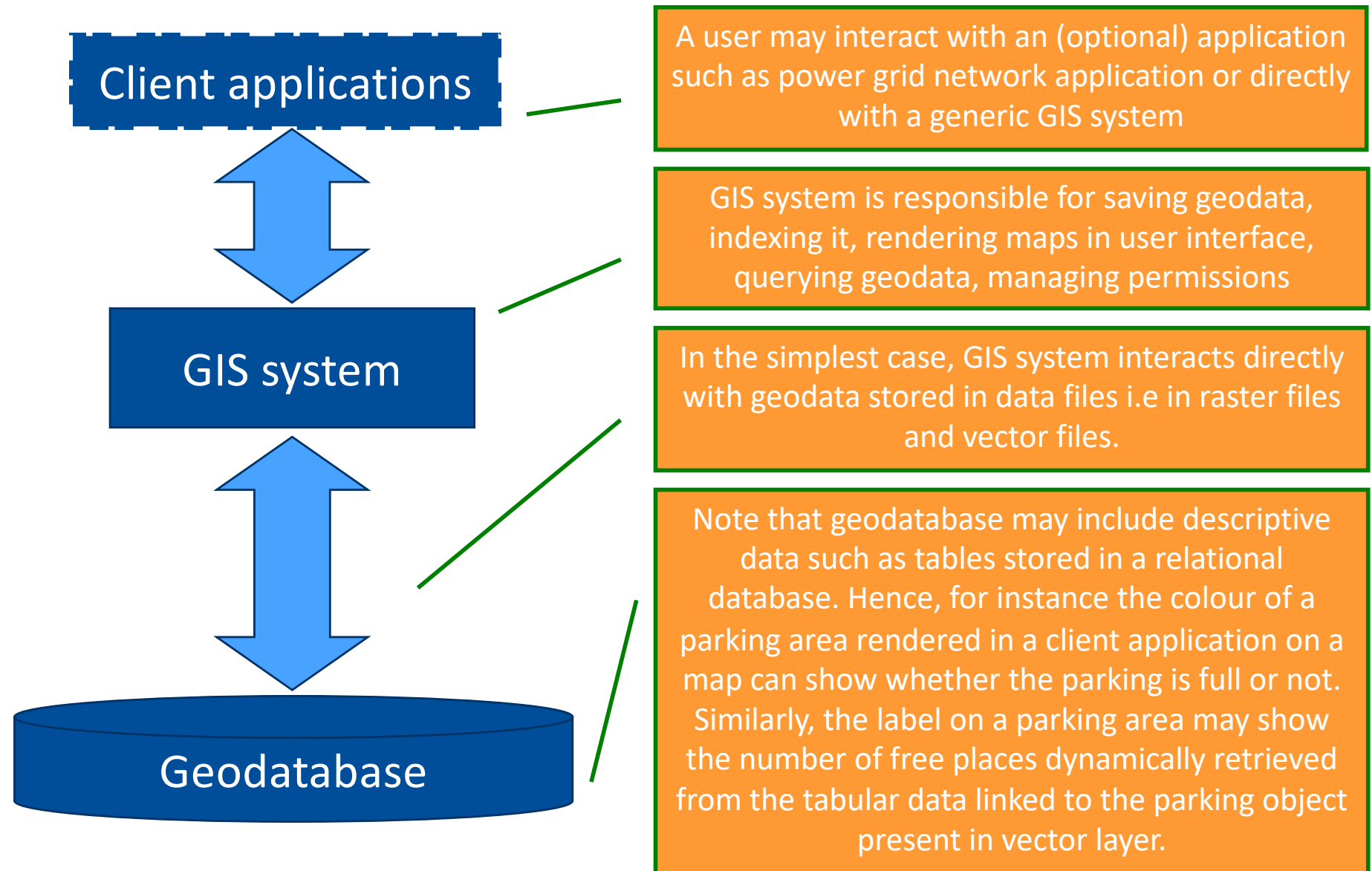
Raster layers and vector layers

- Raster layers (warstwy rastrowe) include:
 - Scanned paper-based maps,
 - Orthophotos (aerial photos), satellite images
- Raster layers are stored as bitmaps linked to the geocoordinates of the region which a tile of a map represents. In addition, the geodatabase may include preprocessed raster images to be shown at non-default scales
- Vector layers (warstwy wektorowe). A vector layer is a collection of objects, each having its shape, geographical coordinates and descriptive attributes such as street name, city name or valve diameter and number.
- Unlike raster layers, vector layers:
 - make sophisticated queries possible
 - can be displayed at an arbitrary scale with equally high quality

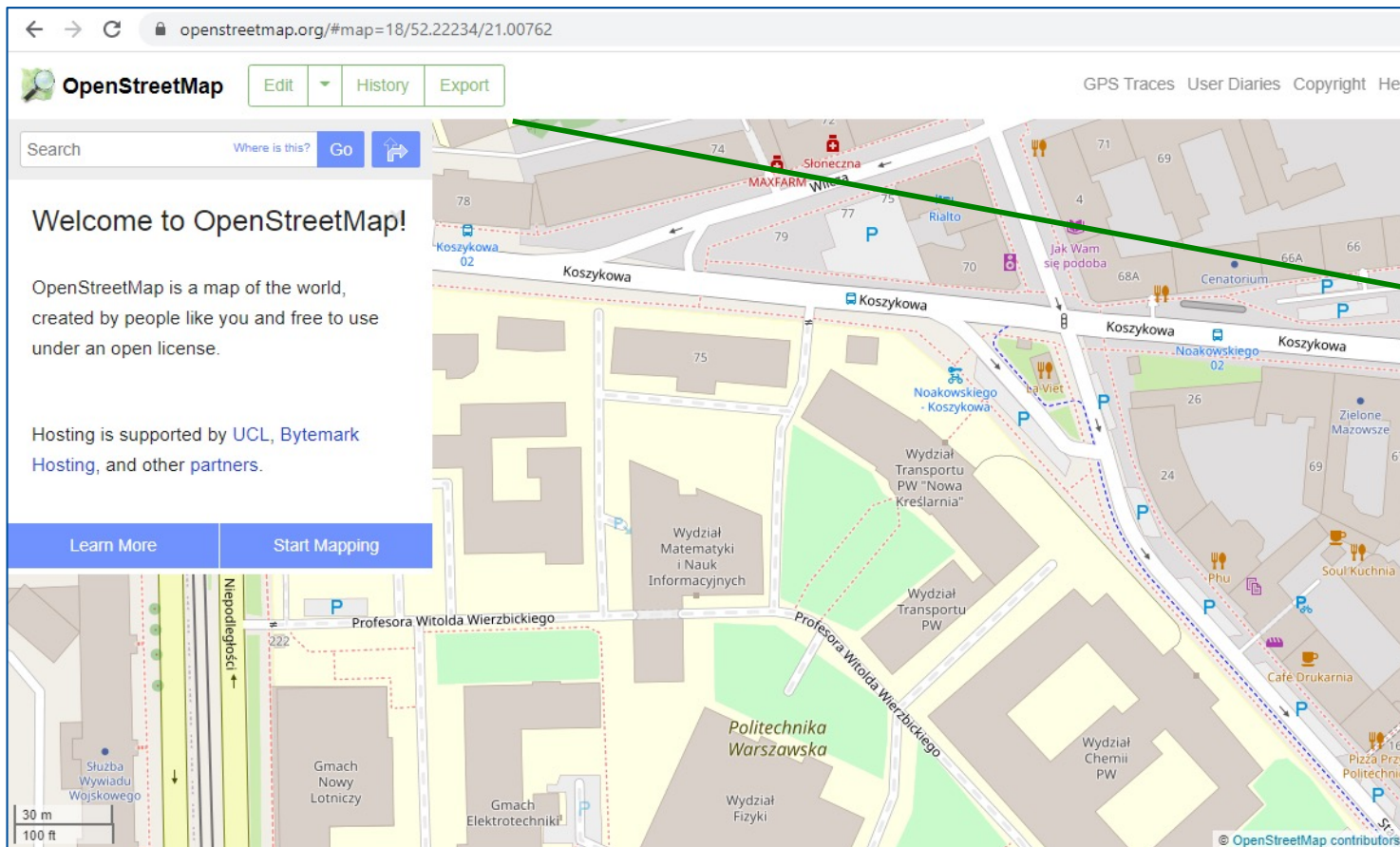
GIS

- Geographical Information System (GIS) (System Informacji przestrzennej) *is a collection of computer hardware, software and geographic data for capturing, managing, analyzing and displaying every form of geographically referenced information, often called spatial data* (source: <https://www.gartner.com/en/information-technology/glossary/geographic-information-systems-gis>)
- GIS is used to manage geodatabases and generate maps out of raster and vector layers
- Top providers include:
 - ESRI
 - Autodesk Inc.
 - Bentley Systems Inc.
- Business applications include:
 - Utilities (gas, water supply (sieć wodociągowa), sewer systems (systemy kanalizacyjne), hydrology, power grid (sieć energetyczna), telecoms, district heating systems (systemy ciepłownicze)
 - Marketing analysis in large organisations
 - Service optimisation in large/medium-scale organisations
 - Government incl. military solutions

The architecture of a GIS system



GIS in practice



While geodata is rendered as a map, the content of the geodatabase can be also exported (here: as XML)

Source: <https://www.openstreetmap.org/>

Note challenges faced by GIS developers such as online generation of map images based on geodata, avoiding overlapping labels, mixing raster and vector data, scalable symbols and much more. This is why one of leading commercial or open sources GIS software packages rather than in-house applications is used when a GIS is needed.

Vector layers (warstwy wektorowe)

- Each vector layer, depending on the GIS system contains objects of the same or different shapes.
- Typical categories of vector layers are:
 - Polyline (linia łamana)
 - Point (punkt)
 - Polygon (wielobok)
- Problems:
 - Different coordinate systems (układy współrzędnych) were developed in the past and are still in use
 - In Poland 5+ standard coordinate systems exist
 - ESRI ArcGIS platform supports 4000+ coordinate systems

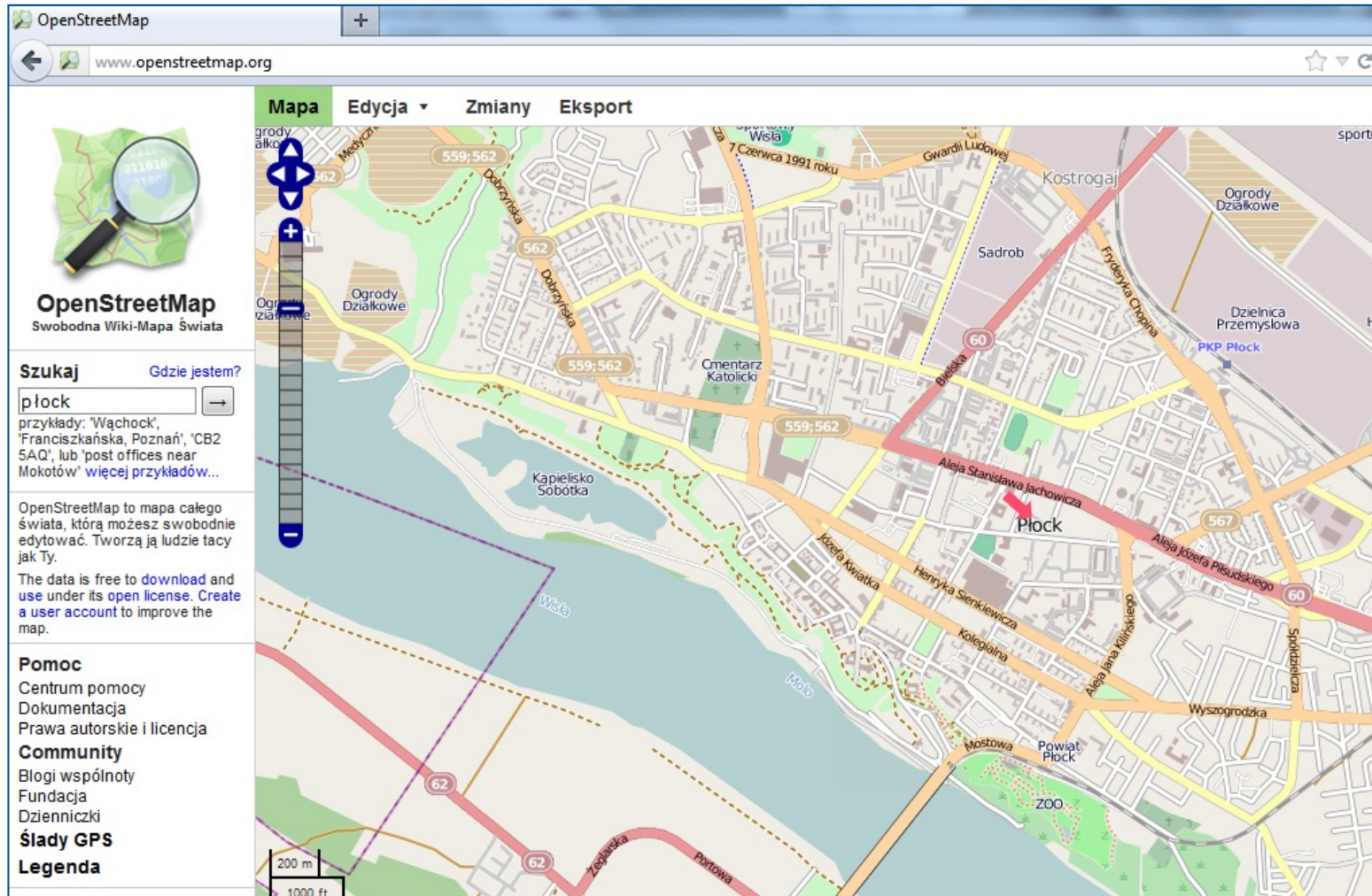
Business applications

- GIS platforms used by business entities go beyond simple solutions of popular GIS platforms
- As an example a GIS system for waterworks company would have:
 - Possibly hundreds of very detailed layers in the system such as: valves, fire hydrants, water supply pipes, water purification plants, pressure reducing valves, drainage, chambers,...
 - Additional tools e.g. for simulating the impact of broken connections, outage, selecting closest valves to cut off leaking pipes, buildings not supplied with water in the case of cutting of a network section,....

Maps – data storage

- Simple solution:
 - collection of files e.g. ESRI SHAPE files, RDL/DGN (CAD format), DXF (CAD format),...
 - drawback:
 - limited editing capabilities: one file can be edited by one user only
 - complicated management: thousands of files used in some installations
- Enterprise solution:
 - Geodata is stored in a standard RDBMS (Oracle, Ms SQL Server,...)
 - or managed by a Spatial extension of RDBMS
 - Many users can potentially edit the same map layer at the same time

GIS – free and commercial data sets



Source: <http://www.openstreetmap.org/>

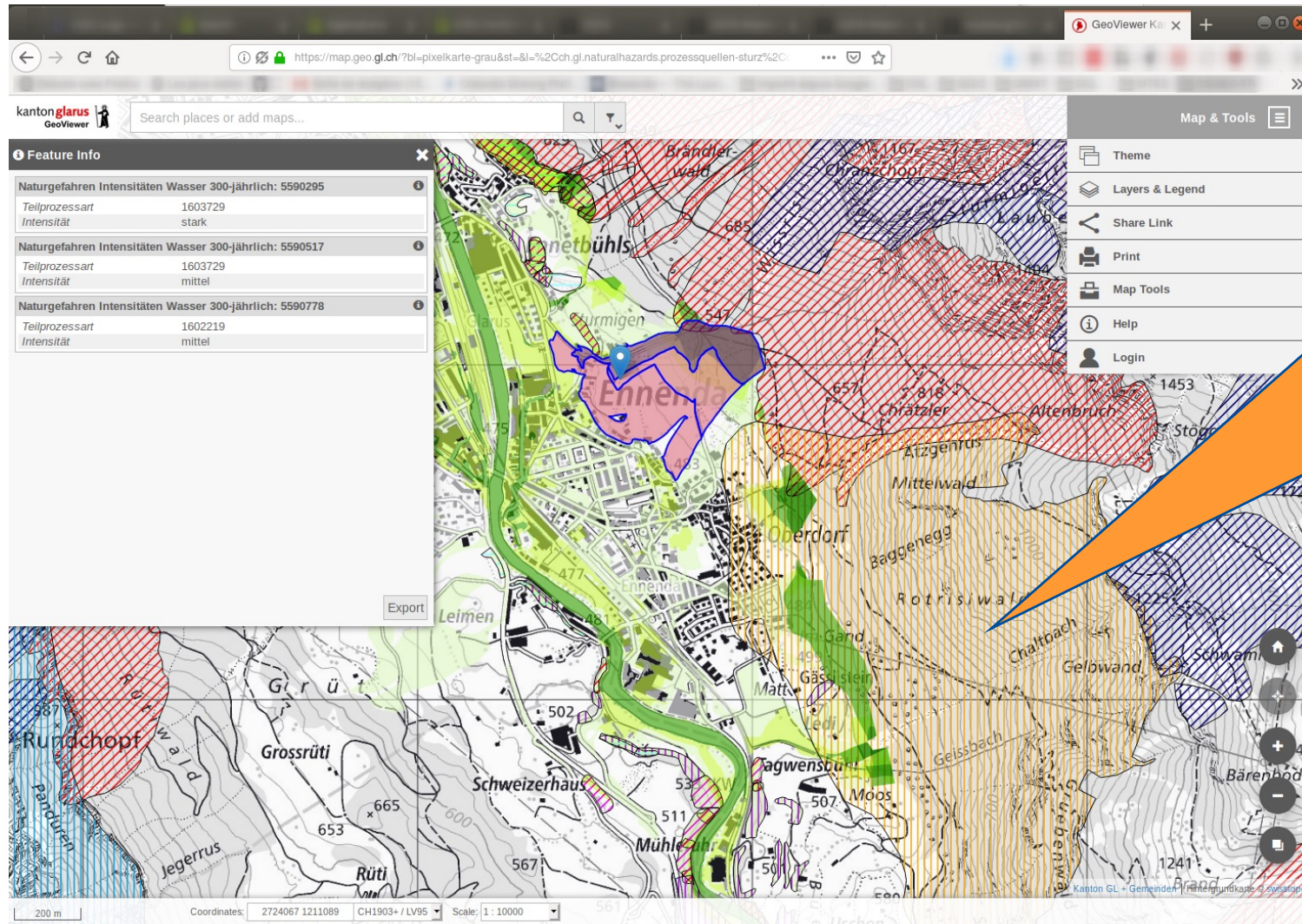
Some geodata are publicly available. This includes OpenStreetMap and open data made available by government and cities. Many data sources are expensive to get. This includes satellite images, maps created out of historical data of the enterprises, data purchased from other organisations.

Non-commercial GIS platforms

- One of the most successful free GIS platforms is QGIS (<https://qgis.org>) , which is an open source platform
- Similarly to other complex offerings, QGIS includes:
 - QGIS Desktop – used to edit, analyse and publish geodata
 - QGIS Server – used to publish layers through network services such as WMS and WFS services
 - QGIS Web client – used to publish GIS projects on the web i.e. make them available via web browsers

Having layers is not the same as having a GIS project. A GIS project includes references to individual layers to comprise on a map, but also complex settings defining symbols to be used (possibly different at different scales), labels to be placed on a map for vector layers (e.g. street names or house numbers), which layers should be shown at which scales, layers shown at request only; the width, style and colour of line, polygon, and point symbols, which can be also data dependent e.g. to make line width reflect pipe diameter.

Sample QGIS visualisation



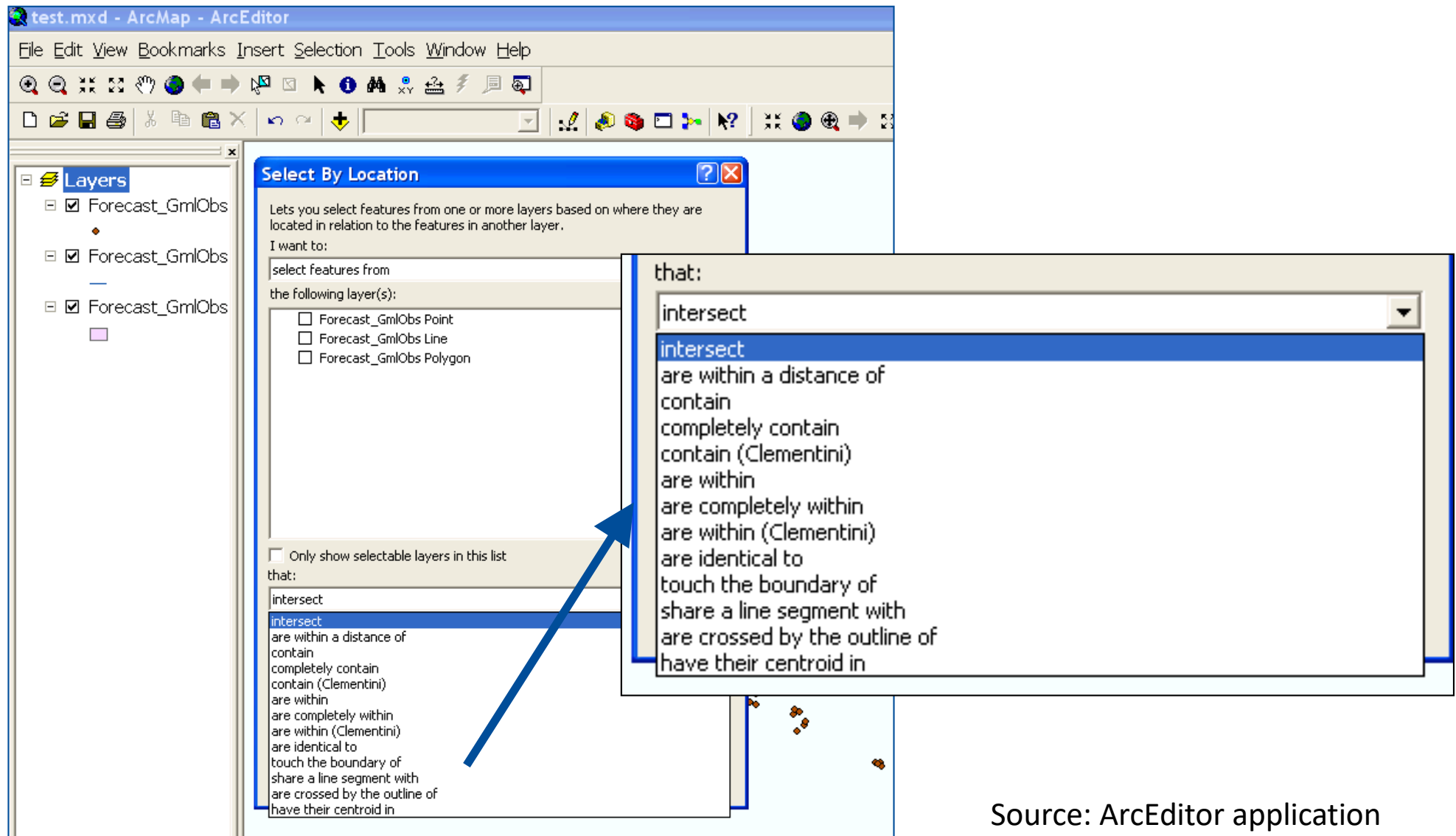
Note that similarly to other GIS tools, a map image is dynamically rendered on the fly based on possibly gigabytes of data of raster and vector layers included in the GIS project.

Source: <https://qgis.org/en/site/about/features.html>

Beyond map rendering: distinctive features of GIS

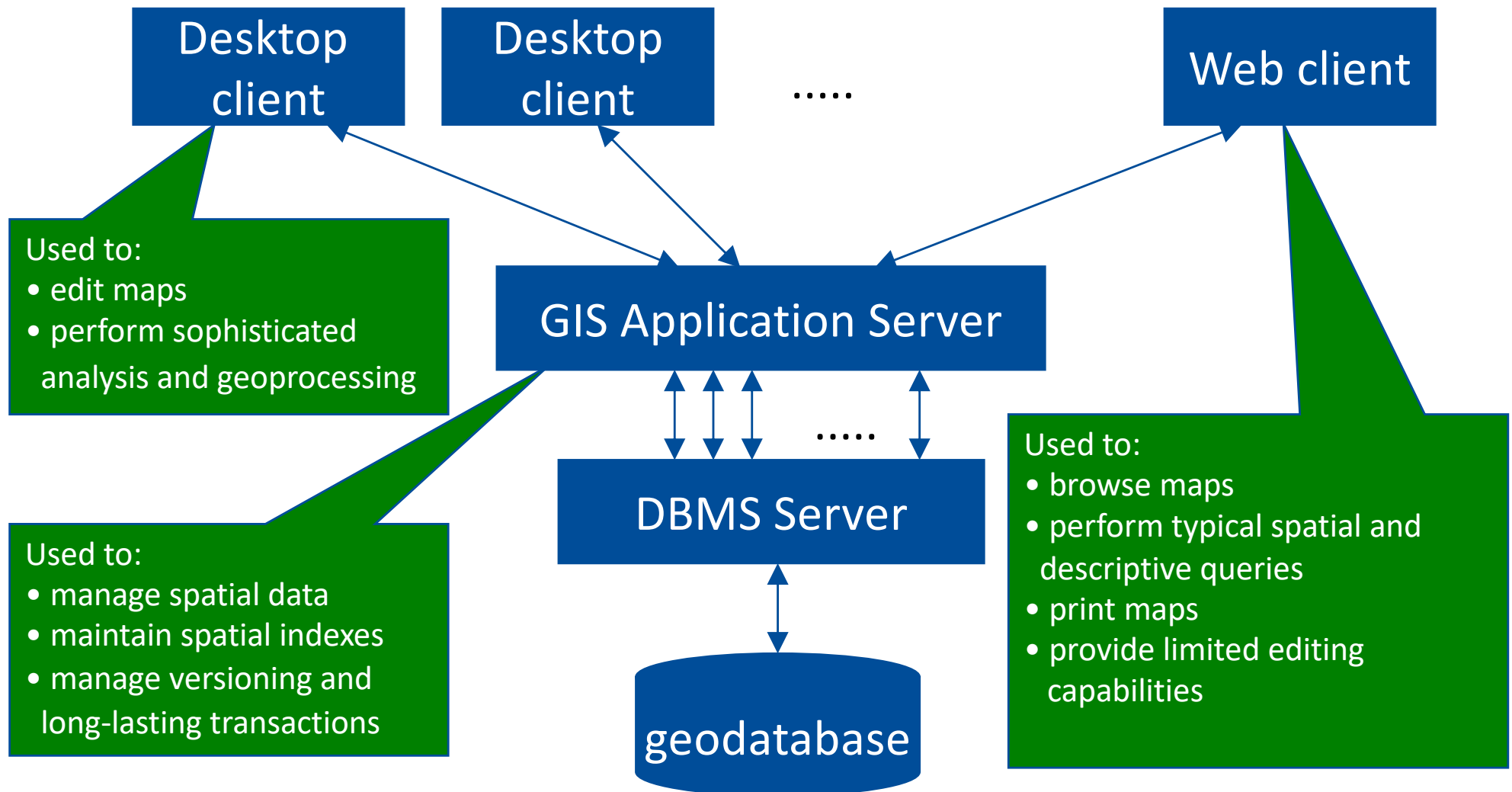
- Every map object of a vector layer is composed of spatial and descriptive data:
 - Shape (e.g. collection of points),
 - Location,
 - Attributes e.g. pipe material or the name of the river
- Hence, queries can be done both in view of spatial information (all the buildings within the distance of 100 meters from the river) and descriptive data (that are made of concrete)

Queries – examples

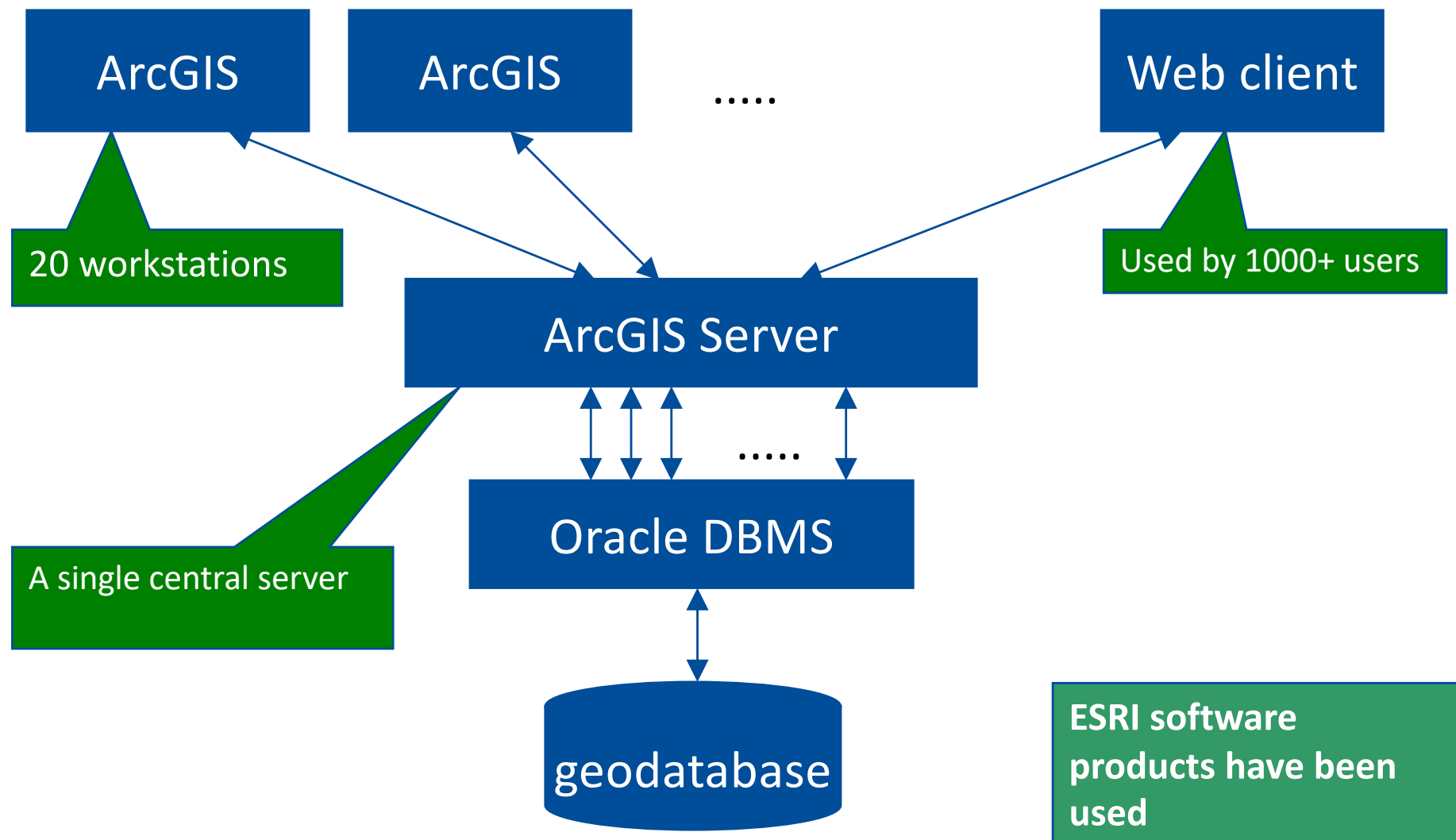


Source: ArcEditor application

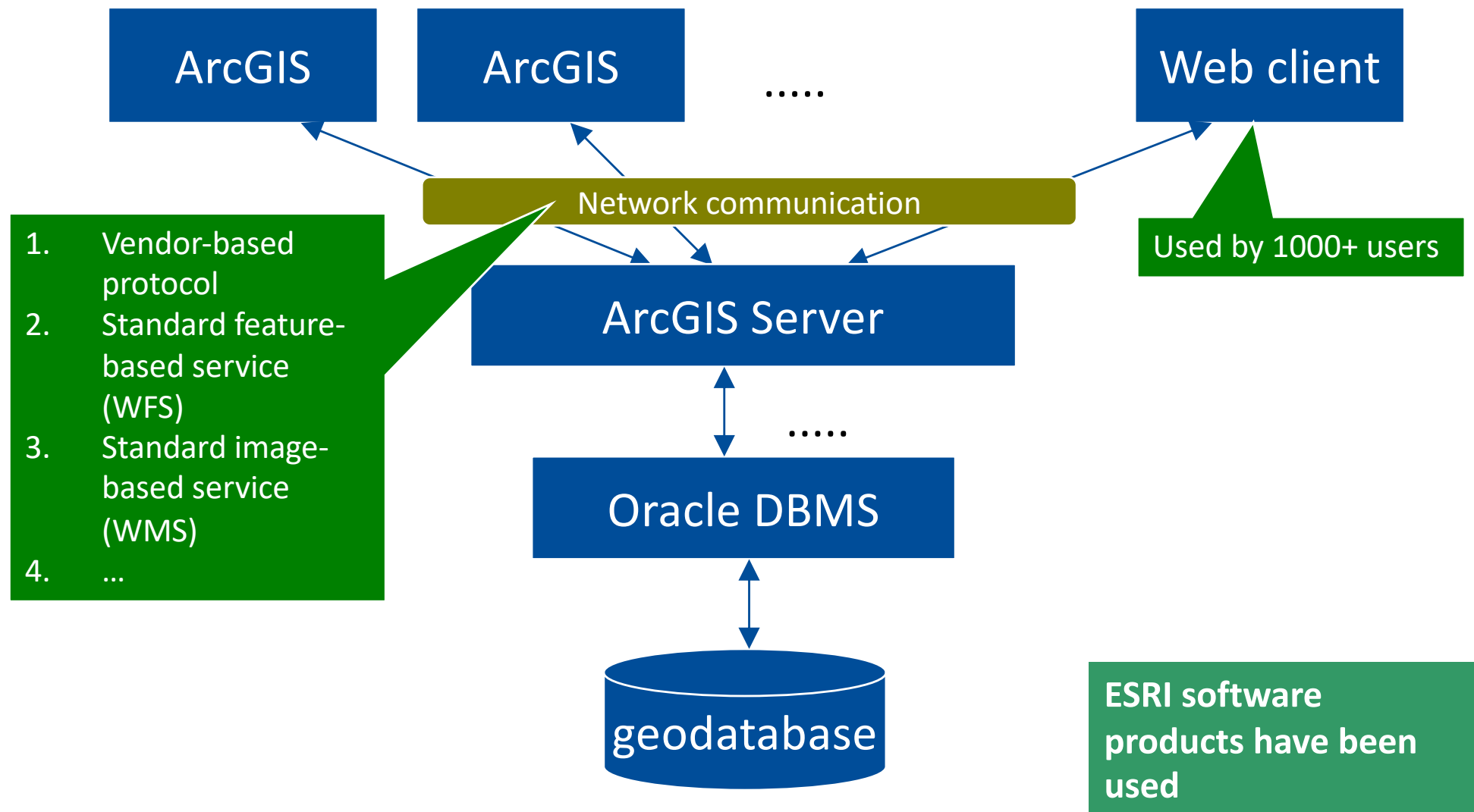
Sample system architecture



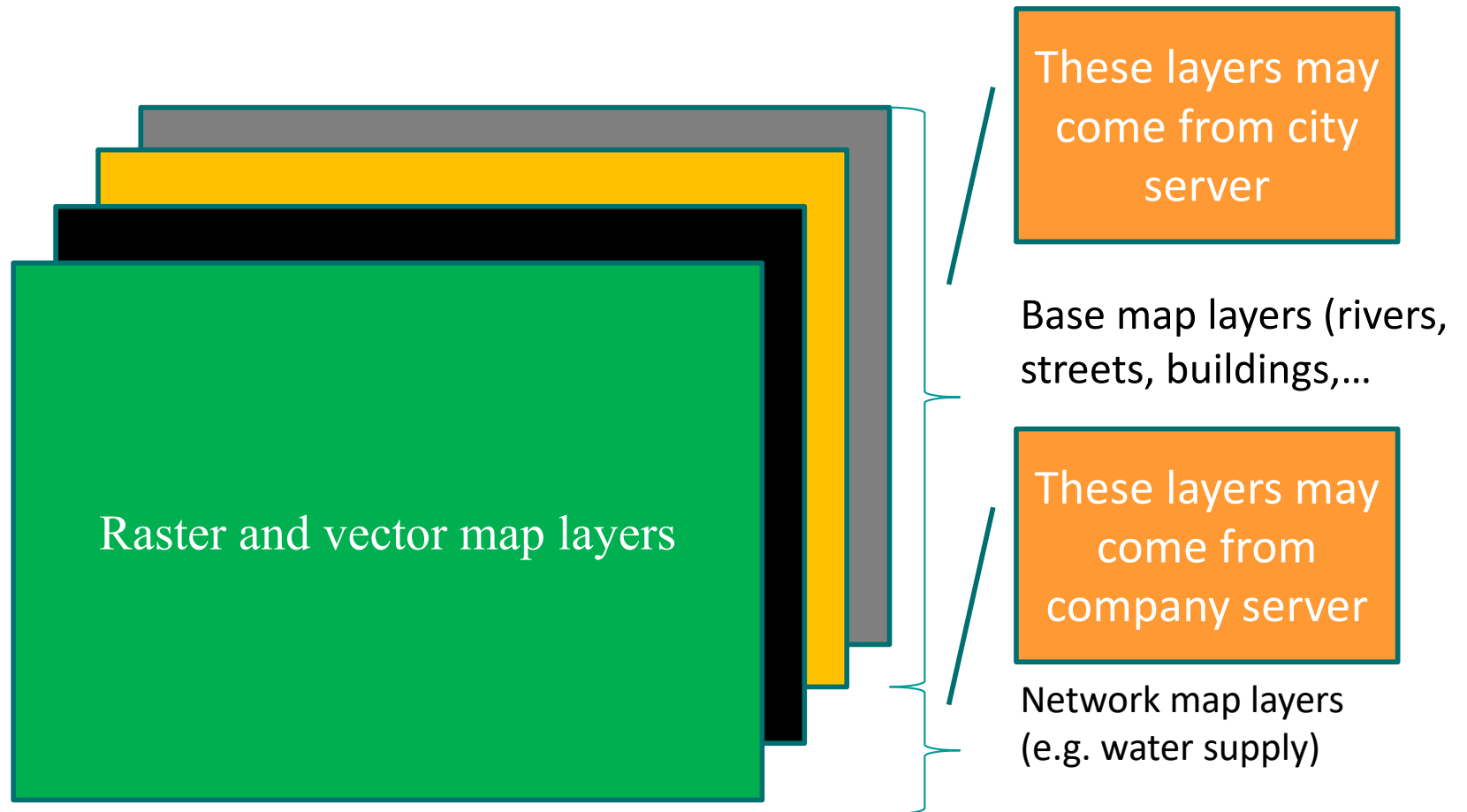
Sample system architecture – case study



Communication standards

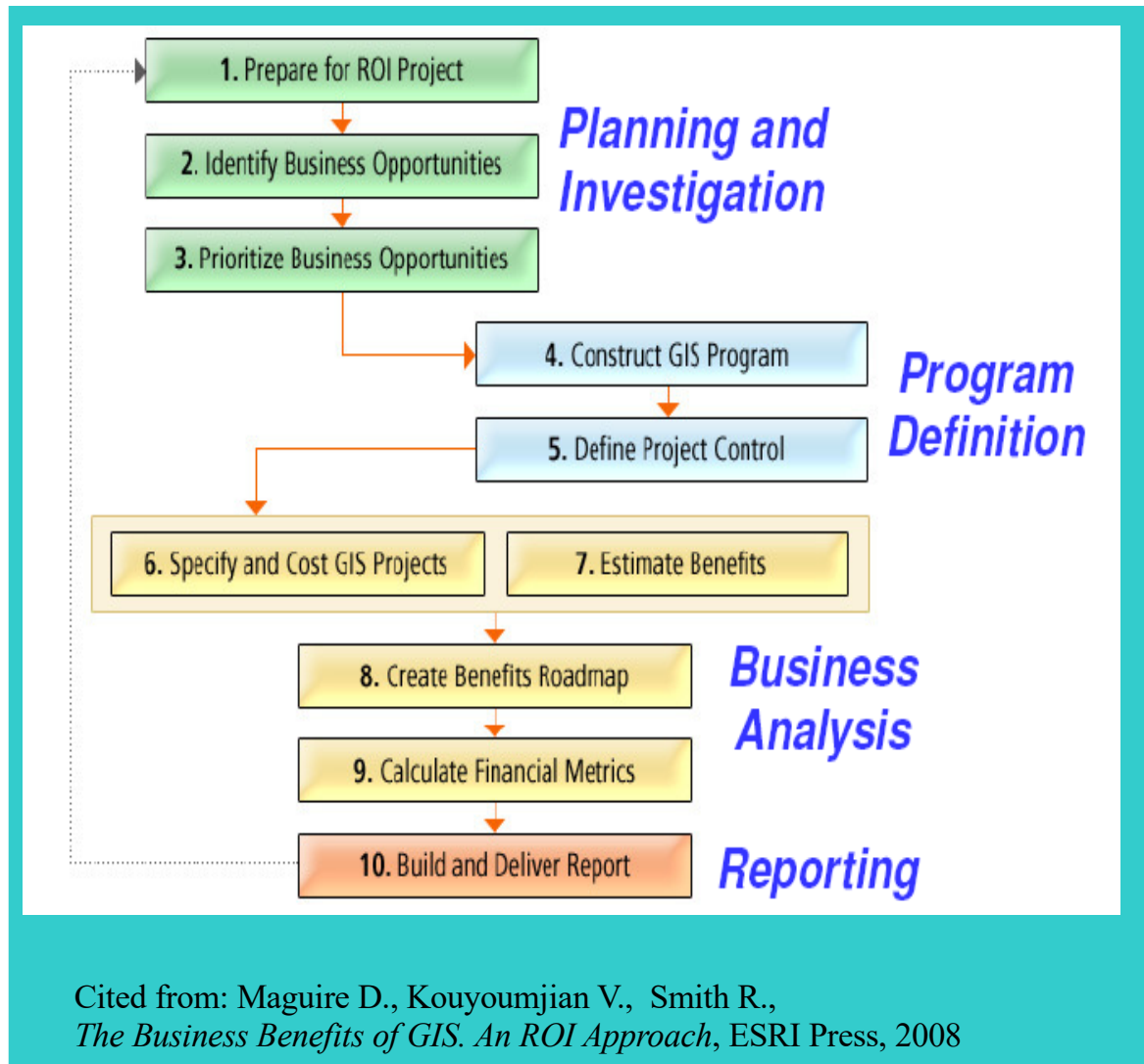


GIS – data layers



WMS (Web Map Service) can be used to integrate on-the-fly map layers coming from different servers of different organisations to render map images based on most recent data from different organisations.

GIS Program development

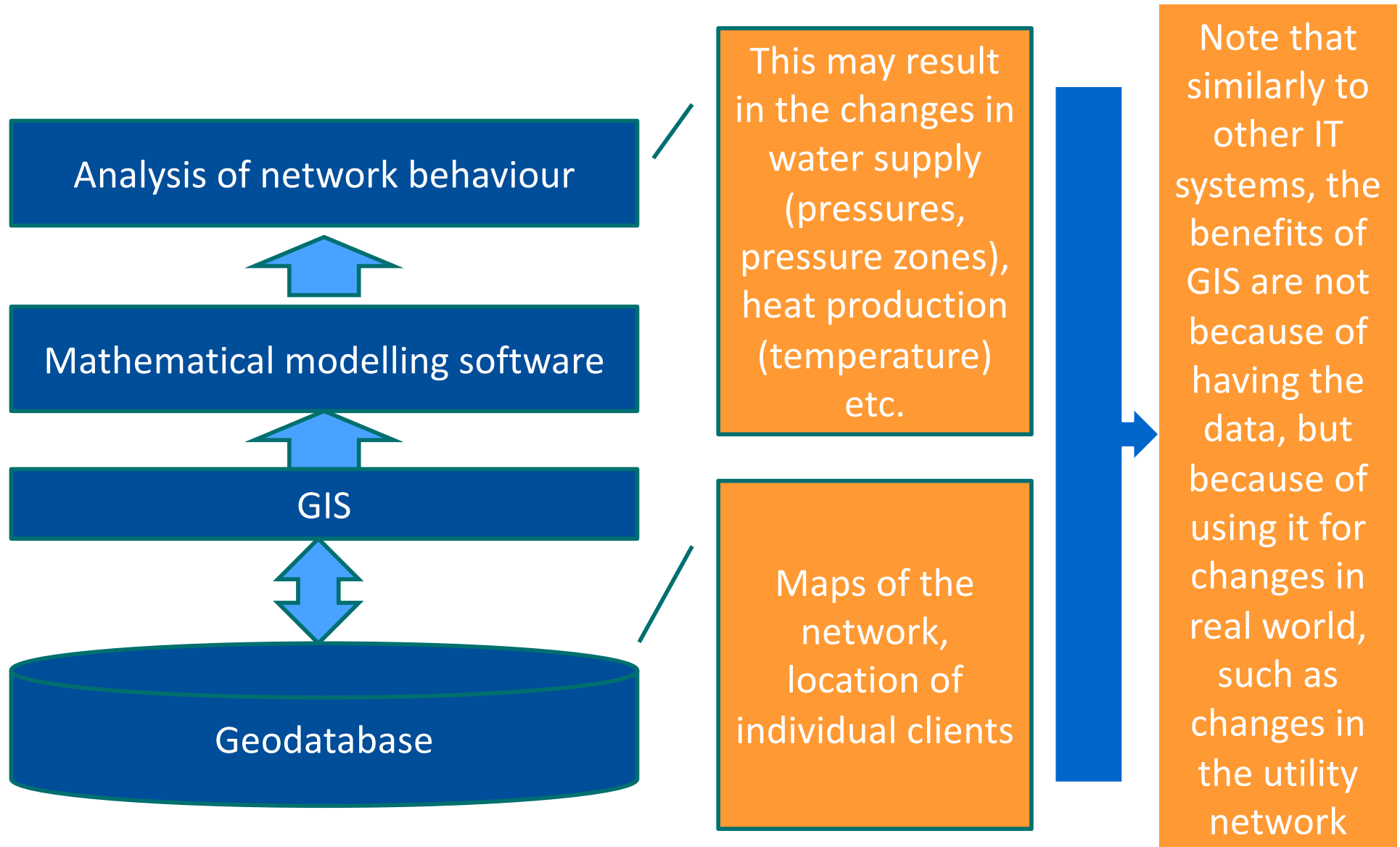


- Map acquisition is frequently an expensive process as it may include:
 - Making aerial photos,
 - Scanning paper-based maps,
 - Calibration,
 - Digitisation,
 - Geocoding
- ROI calculation for a GIS project is not an easy task
- Thus, complete methodology aimed at GIS program planning has been developed

ROI in GIS – utility enterprises

Improved decision making	
RESULT1	Better new investment and maintenance work planning thanks to detailed knowledge on existing network (e.g. water supply network)
RESULT2	Heat/water production /sewage disposal optimisation thanks to mathematical modelling
RESULT3	Decisions on connecting new clients made in view of the capacity of existing network systems
Improved business processes	
RESULT1	Time-consuming data gathering eliminated
RESULT2	Immediate reports taking into account all the spatial and descriptive features possible (e.g. all the water supply pipes with diameter 100mm, located in this area, built in the 1960s)

GIS and network modelling



Summary

- GIS systems are largely needed by organisations which work with assets and persons, which are spatially distributed
- Map interface is much more intuitive in such cases than browsing tabular data of a relational database only
- Furthermore, sophisticated queries referring to spatial data can be answered
- Importantly, location data can be the only data, which makes it possible to identify real-life objects such as valves located underground in park or forest areas
- Furthermore, due to the cost of the construction and maintenance works in the field, GIS systems should be recognised for their potential of reducing excessive costs of these works



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